Question 1 : Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?

Ans- Data quality in ETL pipelines refers to the accuracy, consistency, completeness, and timeliness of data as it moves from sources to a target system. It is a ongoing framework ensuring data reliability for analytics, going beyond simply cleaning errors to encompass proactive validation, structuring, and governance.

It is More Than Just Data Cleaning:

- Proactive vs. Reactive: Data cleaning removes errors (e.g., removing duplicates, fixing nulls). Data quality encompasses the whole process, including preventing issues via data profiling, schema validation, and monitoring pipeline health before data is even loaded.
- Data Structure and Context: Quality involves transforming data to fit business logic, such as data aggregation, data type conversion, or normalization, which is not strictly "cleaning" but restructuring for usability.
- Trust and Governance: High-quality data is not just clean; it is also reliable (available when needed), trusted (data lineage is clear), and compliant with regulations.

Question 2 : Explain why poor data quality leads to misleading dashboards and incorrect decisions.

Ans: It breaks the "garbage in, garbage out" rule, where inaccurate, incomplete, or inconsistent raw data leads to unreliable, misleading dashboards and, consequently, faulty business decisions as ETL processes often consolidate data from multiple sources, they can amplify small errors into systemic failures across the entire organization.

Question 3 : What is duplicate data? Explain three causes in ETL pipelines.

Ans: Duplicate data refers to identical or near-identical appearing multiple times within a dataset. It causes inflated metrics, incorrect reporting, and wasted storage.

Three Causes in ETL Pipelines:

1. **Improper Data Integration (Multiple Sources)**: Merging data from disparate systems that contain overlapping records.
2. **Faulty Job Reruns/Reprocessing**: If an ETL job fails mid-process and is restarted without being "idempotent," it may re-extract and re-load data that was already processed, resulting in doubled records.
3. **Lack of Constraints/Upsert Logic**: When the target database lacks unique constraints or the transformation logic fails to perform an "upsert" (update if exists, insert if new), incoming data is simply appended rather than replacing old data which is causing duplicates

Question 4 : Differentiate between exact, partial, and fuzzy duplicates.

Ans: **Exact Duplicates (Identical):** are identical byte-for-byte.

Logic: Character-by-character comparison (A == B).

Partial Duplicates (Subset Match) : Partial matches occur when some fields are identical, but others are missing, incomplete, or slightly different (often used for data merging)

Fuzzy Duplicates (Approximate Match) : Fuzzy matching, or approximate string matching, identifies records that are "similar enough" to be considered the same, often using algorithms like Levenshtein distance.

Question 5 : Why should data validation be performed during transformation rather than after loading?

Ans: To catch errors early, minimize data corruption in target systems, improve efficiency, and ensure data trustworthiness for immediate analysis.

Question 6 : Explain how business rules help in validating data accuracy. Give an example.

Ans: Business rules enhance data accuracy by establishing logical constraints and standards that ensure information is consistent, valid, and reliable before it is stored.

Example: food& grocery ordering apps.

Question 7 : Write an SQL query on to list all duplicate keys and their counts using the business key (Customer_ID + Product_ID + Txn_Date + Txn_Amount)

```
SELECT   Customer_ID,  Product_ID,  Txn_Date, Txn_Amount,
 COUNT(*) AS DuplicateCount
FROM  Sales_Transactions
GROUP BY Customer_ID, Product_ID, Txn_Date,  Txn_Amount
HAVING  COUNT(*) > 1;
```

Question 8 : Enforcing Referential Integrity Assume the following table: Identify Sales_Transactions.Customer_ID values that violate referential integrity when joined with Customers_Master and write a query to detect such violations.

Ans:

```
SELECT T.Customer_ID, T.Transaction_ID
    FROM Sales_Transactions T
    LEFT JOIN Customers_Master C
    ON T.Customer_ID = C.Customer_ID
     WHERE C.Customer_ID IS NULL;
```